Drude Model (continued)

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- A conductor in an external field $\vec{E} = \vec{E}_0 e^{-i\omega t}$.

  Equation of motion $m\ddot{x} + m\gamma\dot{x} = qE_0 e^{-i\omega t}$ w/ $q = -e$.

- $\vec{j} = \sigma(\omega)\vec{E} = \dfrac{ine^2}{m(\omega+i\gamma)}\vec{E}$; $\cancel{\vec{D} = \varepsilon_0\vec{E} + \vec{P} = \varepsilon(\omega)\vec{E}}$.

  We can relate polarization to current.

$$\left.\begin{aligned} \vec{P} &= \frac{\sum_i q_i \vec{r}_i}{\Delta V} \quad \text{(dipole per unit volume)} \\ \vec{j} &= \frac{\sum_i q_i \vec{v}_i}{\Delta V} \quad (nq\vec{v}) \end{aligned}\right\} \Rightarrow \vec{j} = \frac{\partial \vec{P}}{\partial t} = -i\omega\vec{P} \Rightarrow \vec{P} = \frac{i}{\omega}\vec{j} = \frac{i}{\omega}\sigma(\omega)\vec{E}.$$

So then $\vec{D} = \varepsilon_0\vec{E} + \vec{P} = \left(\varepsilon_0 + \dfrac{i\sigma(\omega)}{\omega}\right)\vec{E} = \varepsilon(\omega)\vec{E}$,

$$\varepsilon(\omega) = \varepsilon_0 + \frac{i\sigma(\omega)}{\omega} \rightarrow \text{fundamental result for linear systems.}$$

$$= \varepsilon_0 - \frac{ne^2}{m\omega(\omega+i\gamma)} = \varepsilon_0\left(1 - \frac{\omega_p^2}{\omega(\omega+i\gamma)}\right), \quad \text{w/ } \omega_p^2 = \frac{ne^2}{m\varepsilon_0} \text{ the plasma frequency.}$$

- How much power is dissipated per unit volume?

  For a single charge, power $= \dfrac{q\vec{E}\Delta\vec{r}}{\Delta t}$. (work per unit time)

  For multiple charges in a volume ΔV, power $\mathcal{P} = \dfrac{\sum_i q_i \Delta\vec{r}_i \vec{E}}{\Delta t\, \Delta V} = \dfrac{\sum_i q_i \Delta\vec{v}_i \vec{E}}{\cancel{\Delta t}\Delta V} = \vec{j}\cdot\vec{E}$.

  per unit volume.

$$\mathcal{P} = \vec{j}\cdot\vec{E}.$$

To find the real part of this quadratic quantity,

$$\mathcal{P} = \mathrm{Re}(\vec{j})\cdot\mathrm{Re}(\vec{E})$$

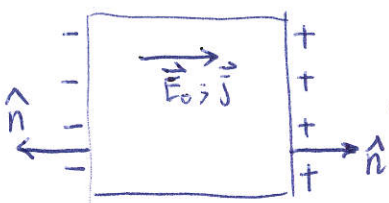
$$= \mathrm{Re}\left(\sigma(\omega)\vec{E}\right)\cdot\mathrm{Re}(\vec{E}) = \mathrm{Re}\left(\sigma(\omega)E_0 e^{-i\omega t}\right)\mathrm{Re}\left(E_0 e^{-i\omega t}\right)$$

$$= \left[\tfrac{1}{2}\left(\sigma(\omega)E_0 e^{-i\omega t} + \sigma^*(\omega)E_0^* e^{i\omega t}\right)\right]\cdot\left[\tfrac{1}{2}\left(E_0 e^{-i\omega t} + E_0^* e^{i\omega t}\right)\right]$$

$$\overset{\text{averaging}}{=} \frac{1}{4}\left[\sigma(\omega) + \sigma^*(\omega)\right]|E_0|^2 = \frac{1}{2}|E_0|^2\,\mathrm{Re}\left[\sigma(\omega)\right].$$

- Now we go back and look at the expression for $\vec{j}$.

$$\vec{j} = \frac{ine^2}{m(\omega+i\gamma)}\vec{E}_{tot},$$ we take $\gamma \ll 1$. When ω decreases and approaches zero, our current blows up. Intuitively, this is just saying in the dc limit, charges accelerate without rest. One way to fix this is to introduce finite size for our system. Then we will have accumulation of charges on the boundaries. We call the surface density of charges Σ.



External field induces current; current induces charge accumulation; accumulated charge induces electric field. We are now at the second step of this process.

- Namely, how to relate Σ to $\vec{j}$? We look at the right boundary now.

$$\Sigma = \vec{P}\cdot\hat{n} = \frac{i}{\omega}\vec{j}\cdot\hat{n} = \frac{i}{\omega}\frac{ine^2\vec{E}_{tot}}{m(\omega+i\gamma)} = -\frac{ne^2\vec{E}_{tot}}{m\omega(\omega+i\gamma)} = -\frac{\varepsilon_0\omega_p^2}{\omega(\omega+i\gamma)}\vec{E}_{tot}$$

- The total electric field.

$$\vec{E}_{tot} = \vec{E}_0 + \vec{E}_{ind} = \vec{E}_0 - \frac{\Sigma}{\varepsilon_0} = \vec{E}_0 + \frac{\omega_p^2}{\omega(\omega+i\gamma)}\vec{E}_{tot}.$$

Solving this, $$\vec{E}_{tot} = \frac{1}{1-\frac{\omega_p^2}{\omega(\omega+i\gamma)}}\vec{E}_0 = \frac{\omega^2+i\omega\gamma}{\omega^2-\omega_p^2+i\omega\gamma}\vec{E}_0$$

when you go to $\omega \to 0$, field inside the conductor will vanish.

So. $$\vec{j} = \frac{ine^2}{m(\omega+i\gamma)}\frac{\omega(\omega+i\gamma)}{(\omega^2-\omega_p^2)+i\omega\gamma}\vec{E}_0$$

$$= \frac{ine^2\omega}{m}\frac{1}{(\omega^2-\omega_p^2)+i\omega\gamma}\vec{E}_0.$$

- Now compare
$$\begin{cases} \vec{j} = \dfrac{ine^2}{m}\dfrac{1}{\omega+i\gamma}\vec{E}_{tot} & \text{response to the local field.} \\[2ex] \vec{j} = \dfrac{ine^2}{m}\dfrac{\omega}{(\omega^2-\omega_p^2)+i\omega\gamma}\vec{E}_0 & \text{response to the applied field.} \end{cases}$$

The second equation has a singularity at $\omega = \omega_p$, plasma frequency.

- Remark. The factor $\frac{1}{(\omega^2-\omega_p^2)+i\omega\gamma}$ has the form of a Lorentzian, (same thing for a driven oscilattor.) The smaller the γ, the sharper the singularity at $\omega=\omega_p$. This singularity corresponds to an eigenmode called plasma wave, which is longitudinal: it polarizes along $\vec{E}$.

(Suppose at some time we stop driving $\vec{E}$, ~~then~~ but we still have Σ there. They still induce current. This longitudinal charge oscillation at ω_p is our plasmon.)